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<h1 align="center"> 📄 Smart Attendance System</h1>

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  <b>Real-Time Face Recognition · Anti-Spoofing Liveness Detection · AI-Powered
Chatbot</b>

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  <a href="https://www.python.org/downloads/"></a>

  <a href="https://opencv.org/"></a>

  <a href="https://streamlit.io/"></a>

  <a href="https://ai.google.dev/"></a>

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An end-to-end, AI-powered attendance solution that leverages ****OpenCV**** for real-time facial recognition and ****anti-spoofing liveness detection****. The system automates the check-in process through a sleek Streamlit dashboard and features an integrated Gemini-powered chatbot for natural language record querying.

🎯 Features

Feature	Description
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 Advanced Face Recognition	High-accuracy matching using LBPH and DNN-based embeddings
 Anti-Spoofing Liveness	Blink-frequency analysis to ensure a live person, not a photo
 AI Attendance Assistant	RAG chatbot powered by Gemini 2.5 Flash + FAISS for natural language queries
 Automated Analytics	Daily/monthly reports with visualizations, exportable to PDF, Excel, and CSV
 Live Enrollment	Register students directly via camera with automated dataset training

Tech Stack

Category	Tools
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Computer Vision	OpenCV, DeepFace, Imutils
Frontend	Streamlit, Plotly
LLM	Google Gemini 2.5 Flash
Database & Vector Search	SQLite3, FAISS
Data Processing	Pandas, NumPy, Scikit-Learn

Project Structure

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Smart-Attendance-System

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- |— main.py # Entry point — initializes dirs & launches app
- |— dashboard.py # Streamlit UI (live feed, reports, controls)
- |— face_recognition_system.py # Core OpenCV + liveness detection logic
- |— chatbot.py # Gemini + FAISS RAG chatbot
- |— database.py # SQLite handling & report generation
- |— embedding_pipeline.py # Face embeddings + ML (SVM) training
- |— requirements.txt # Python dependencies

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- └─ data/ # Auto-generated at runtime
 - |— attendance.db # SQLite attendance database
 - |— models/ # Trained LBPH & SVM models
 - └─ embeddings/ # Facial embedding vectors

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Quick Start

Prerequisites

- Python 3.9+
- A webcam
- A [Google Gemini API key](<https://aistudio.google.com/app/apikey>)

1. Clone the Repository

```
```bash  

git clone https://github.com/Zerodistracton-max/Smart-Attendance-with-Face-Liveness-
Detection.git

cd Smart-Attendance-with-Face-Liveness-Detection
```
```

2. Install Dependencies

```
```bash  

pip install -r requirements.txt
```
```

3. Environment Setup

Create a `.env` file in the root directory and add your credentials:

```
```env  

GOOGLE_API_KEY=your_gemini_api_key_here
ATTENDANCE_DB_PATH=data/attendance.db
```
```

4. Run the Application

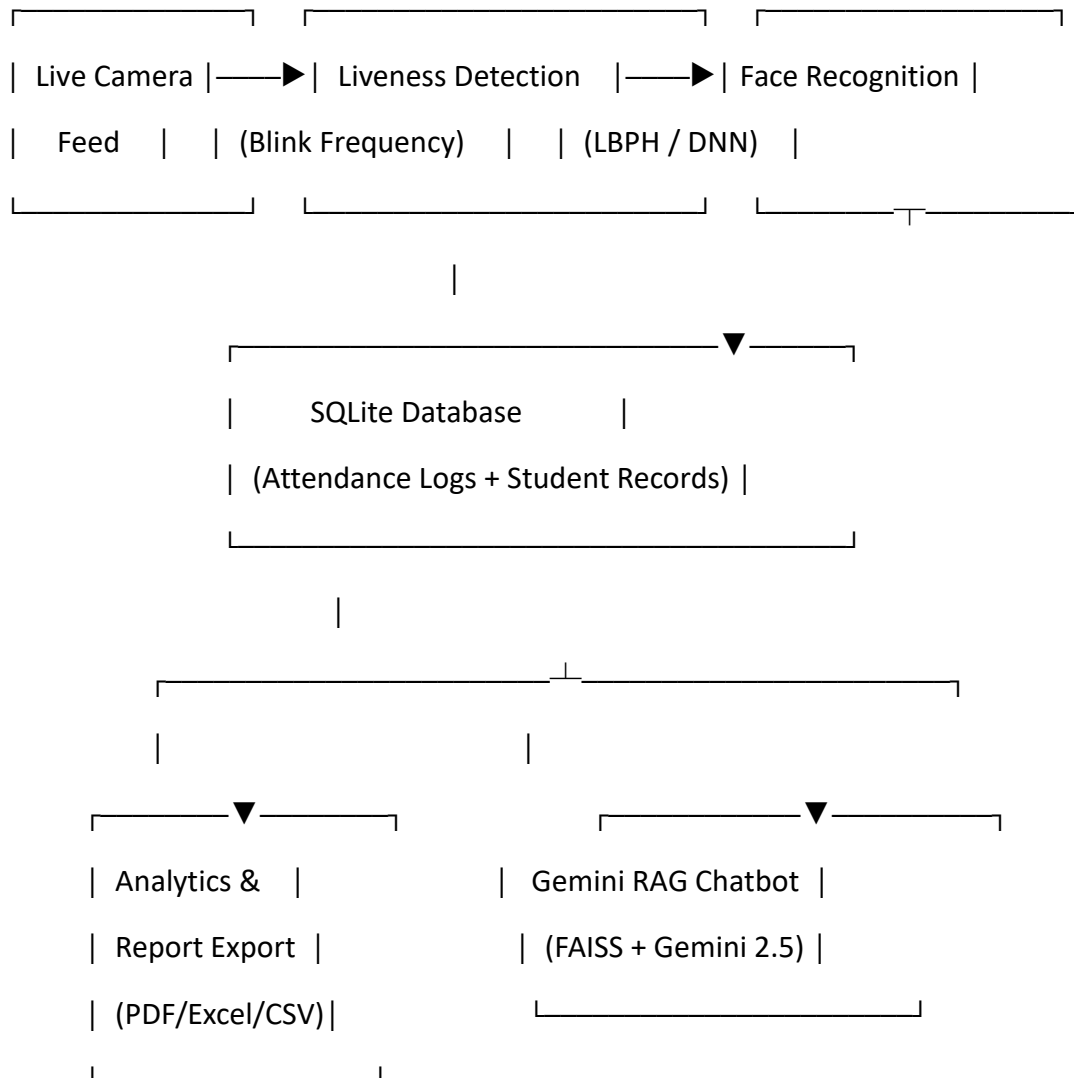
```
```bash  

python main.py
```
```

> The system will automatically initialize all required directories and launch the Streamlit dashboard in your browser.

🖥️ How It Works

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1. ****Live Feed**** — The camera captures a real-time video stream.
2. ****Liveness Check**** — Blink detection confirms the subject is a live person.
3. ****Recognition**** — LBPH or DNN embeddings identify the student.
4. ****Logging**** — A timestamped record is saved to SQLite.
5. ****Analytics**** — Reports are auto-generated and available for export.
6. ****Chatbot**** — Ask questions in plain English like *"Who was absent on Monday?"*

🗣️ AI Chatbot Examples

The Gemini-powered RAG assistant can answer natural language queries over your attendance database:

- > 🗣️ *"Was Rahul present yesterday?"**
- > 🗣️ *"How many students were absent last week?"**
- > 🗣️ *"Show me the attendance summary for March."**
- > 🗣️ *"Who has the lowest attendance this month?"**

📊 Reports & Analytics

- **Daily Summary** — Per-day attendance breakdown
- **Monthly Report** — Aggregated monthly statistics with charts
- **Export Formats** — PDF, Excel (`.xlsx`), and CSV
- **Visualizations** — Bar charts and trend graphs via Plotly

🔧 Configuration

| Variable | Description | Default |
|----------|-------------|---------|
|----------|-------------|---------|

| | | |
|-----|-----|-----|
| --- | --- | --- |
|-----|-----|-----|

| `GOOGLE\_API\_KEY` | Your Gemini API key | **(required)** |
| `ATTENDANCE\_DB\_PATH` | Path to SQLite database | `data/attendance.db` |

🤝 Contributing

Contributions are welcome! Here's how to get started:

1. **Fork** the repository
2. **Create** a feature branch: `git checkout -b feature/your-feature-name`
3. **Commit** your changes: `git commit -m 'Add some feature'`
4. **Push** to the branch: `git push origin feature/your-feature-name`
5. **Open** a Pull Request

Feel free to open an [issue](<https://github.com/Zerodistracton-max/Smart-Attendance-with-Face-Liveness-Detection/issues>) for bug reports, feature requests, or questions.

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Made by Suryansh Sapehia

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